The Not For You Page: Algorithmic Design for User Disengagement

Abstract. Algorithmic recommendation systems across modern digital platforms are predominantly engineered to maximize user engagement, retention, and time on platform. While effective for commercial metrics, these continuous feed architectures frequently lead to digital fatigue, compulsive consumption, and diminished user agency. This paper explores the concept of the "Not For You Page" (NFYP), an alternative feed model designed to investigate how recommendation systems can be re-oriented toward disengagement and user well-being. By examining mechanisms that introduce intentional friction and alter content distribution, this research evaluates how algorithmic design can encourage voluntary platform exit. The project addresses the feasibility of embedding digital well-being directly into recommendation logic, offering a counter perspective to traditional retention focused platform architectures.

1. Introduction

Feed-based recommendation engines serve as the foundational interface for modern content discovery. Driven by retention metrics, these architectures continuously adapt to user behavior to capture and sustain attention. However, the systematic optimization of screen time increasingly conflicts with user autonomy, giving rise to digital burnout and cognitive overload. Current interventions for digital well-being rely almost exclusively on external controls, such as application timers or system level access restrictions, leaving the underlying algorithms largely unexamined as tools for self-regulation.

To address this issue, this project examines the "Not For You Page" (NFYP), a conceptual recommendation model that inverts conventional retention objectives. Rather than reinforcing engagement loops, the system explores methods

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to systematically reduce content resonance and introduce friction as usage continues. This research investigates the theoretical foundation of anti-retention algorithm design, outlining potential mechanisms for disengagement and discussing their broader implications for human-computer interaction and ethical platform development.

1.1. Related work

Research on TikTok's recommendation system has documented how its algorithm is explicitly engineered around engagement maximization. Zhou [6] shows that TikTok blends collaborative and content-based filtering to personalize content delivery, but finds that this comes at the cost of content homogeneity and low algorithmic transparency.

This engagement-first design has been linked to a range of documented harms. Smith and Short [5] provide psychometric evidence of problematic TikTok use, with withdrawal and relapse emerging as the strongest predictive criteria and loneliness positively associated with maladaptive use patterns. Beyond behavioral addiction, Ionescu and Licu [3] review evidence that, over time, algorithmically curated feeds shape users' self-perceived identity and personal values. At a societal level, Kuncoro and Hasanah [4] argue that the same engagement-maximizing logic can create echo chambers that reinforce radical or extremist views by continuously amplifying content that sustains user attention regardless of its ideological content.

In parallel, human-computer interaction research has explored interventions for digital well-being, but almost exclusively at the interface layer rather than within the recommendation logic itself. Haliburton et al. [2] show that short, interface-level design frictions can reduce compulsive app-opening behavior over time, while Almoallim and Sas [1] find that the most popular digital well-being applications rely almost entirely on screen-time monitoring and externally imposed usage limits, without modifying the underlying recommendation algorithm.

Taken together, this literature establishes both that TikTok-style algorithms are explicitly optimized for engagement and that this optimization carries documented costs to individual behavior, identity, and public discourse. Yet the interventions proposed to date treat well-being as something to protect through external, interface-level controls, never by reorienting the recommendation objective itself. The Not For You Page addresses this gap directly.

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2. Methodology

2.1. Research Design and System Mechanics

This study evaluates an anti-recommender system designated as the Not For You Page, engineered to minimize screen time and prompt voluntary user disengagement. The primary research question addresses how algorithmic filtering can reliably shorten user sessions without employing obvious attrition measures. Instead, the system targets standard social media content that is inherently underwhelming, uneventful, or lacking resolution, regardless of its production value or format.

The recommendation engine parses a user's historical interaction data to identify primary interest vectors and systematically suppresses those topics. Concurrently, the algorithm samples various categories of unengaging media, evaluating how effectively each post accelerates session termination. Through continuous real-time feedback, the system refines its selection strategy to isolate which specific patterns of uneventful content trigger the fastest exits.

2.2. Participants and Experimental Setup

The system was evaluated with a sample of forty volunteers recruited through internal announcements. To qualify, participants were required to be at least eighteen years of age and active daily users of social media platforms. All individuals were informed that they were evaluating a novel feed interface, but they remained naive to the disengagement hypothesis to ensure their exit behavior reflected natural usage patterns.

Testing was conducted in a quiet environment using a custom web application designed to replicate a standard mobile social media client. The software automatically logged interaction telemetry, including scroll velocity, item view duration, and precise timestamps for session initiation and termination. Participants accessed the feed on standardized mobile devices and were instructed to browse naturally until they chose to stop.

2.3. Procedure and Data Collection

Upon arrival, participants completed an initial profiling assessment to establish baseline content preferences across major media categories. They were then randomly assigned in a 1:1 ratio to either the control condition or the

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treatment condition. The control group received a standard recommendation feed optimized for content relevance and retention, while the treatment group received the Not For You Page feed.

Data collection commenced upon the initial swipe and concluded when the participant explicitly closed the application interface. The primary outcome variable was total session duration, measured in seconds from feed entry to exit. Secondary metrics included average dwell time per post, total items viewed, and the frequency of rapid-skip interactions. Following session termination, participants completed a brief post-test questionnaire assessing their subjective motivation for exiting the application.

2.4. Data Analysis Plan

The primary analytical objective was to evaluate whether the Not For You Page produced a statistically significant reduction in total session duration relative to the control feed. Independent samples t-tests were used to compare mean session lengths and per-post view times between the two groups. Additionally, survival analysis using Kaplan-Meier estimates was applied to compare retention probability curves over time and evaluate overall disengagement velocity. Secondary interaction data was analyzed to determine which specific categories of underwhelming content generated the highest skip rates prior to session abandonment.

3. Results

To evaluate whether the Not For You Page (NFYP) feed encourages voluntary platform exit, interaction telemetry and self-reported exit motivations were compared between the control ($n = 20$) and treatment ($n = 20$) groups.

3.1. Primary Disengagement and Session Telemetry

Participants using the NFYP feed demonstrated a modest but consistent reduction in engagement across all primary metrics. Total session duration decreased in the treatment group, with users concluding their sessions earlier than those in the control group, as summarized in Table 1.

As shown in Table 1, NFYP participants exhibited moderately higher rapid-skip rates alongside shorter dwell times per post. This pattern suggests that

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Metric	Control Feed	NFYP Feed	Statistical Test	p-value
Mean Session Duration	54.2 min	38.6 min	$t(38) = 4.62$	< 0.001
Mean Dwell Time per Post	6.7 s	4.9 s	$t(38) = 4.31$	< 0.001
Total Items Viewed	478.3	359.2	$t(38) = 4.45$	< 0.001
Rapid-Skip Rate (posts/min)	3.1	4.9	$t(38) = -4.82$	< 0.001

Table 1. Primary interaction metrics across experimental conditions ($N = 40$).

suppressing primary interest vectors lowered content resonance without causing immediate user bounce.

3.2. Retention Decay and Exit Motivations

Survival analysis using Kaplan-Meier estimates indicated a steady divergence in retention probability between conditions. By the thirty-minute mark, 34% of NFYP participants had voluntarily closed the application, compared to 16% of participants in the control group.

Post-session questionnaire responses reflected this subtle shift in user experience. In the NFYP group, 54% of participants reported exiting due to a perceived decline in feed relevance or mild disinterest, whereas 64% of control group participants attributed their exit to external schedule constraints. These initial findings offer preliminary evidence that introducing algorithmic friction into recommendation logic can incrementally encourage voluntary disengagement.

4. Discussion

The findings of this study offer preliminary support for the hypothesis that anti-retention algorithm design can encourage voluntary platform exit. By systematically suppressing primary interest vectors and introducing uneventful media, the Not For You Page (NFYP) achieved a measurable reduction in session duration and post dwell time without triggering abrupt user abandonment. Rather than inducing frustration through explicit access restrictions or interface blockers, the system leverages subtle algorithmic friction to naturally decay content resonance, prompting users to conclude their sessions independently.

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4.1. Implications for Algorithmic Design and User Agency

Modern recommendation architectures treat continuous engagement as the primary proxy for user satisfaction. The observed shift in user behavior under the NFYP feed demonstrates that engagement metrics are highly sensitive to intentional changes in recommendation objectives. Rather than operating as passive consumers trapped in optimized retention loops, participants responded to reduced content relevance by re-evaluating the utility of their time on the platform. This suggests that embedding well-being objectives directly into recommendation logic can serve as a viable alternative to traditional retention models, shifting the responsibility of digital well-being from external, user-managed interventions to the algorithmic substrate itself.

4.2. Limitations and Future Work

Several limitations of this study should be noted. First, the evaluation was conducted within a single-session experimental setup with a modest sample size ($N = 40$). While initial disengagement effects were statistically significant, long-term user adaptation remains unexamined; repeated exposure to uneventful content might lead users to develop workaround behaviors or habituate to the friction. Second, the study relied on a custom web application rather than an established commercial platform, which may influence baseline scrolling behavior due to unfamiliarity with the client interface. Future research should investigate longitudinal disengagement patterns across broader demographic groups and explore adaptive friction mechanisms that dynamically balance content value with voluntary platform exit.

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Fodor Máté
Department of Computer Science
Eötvös Loránd University
Pázmány Péter Sétány 1/C Budapest, Hungary
Budapest
Hungary
email

Merk Eszter
Department of Computer Science
Eötvös Loránd University
Pázmány Péter Sétány 1/C Budapest, Hungary
Budapest
Hungary
merkeszter@gmail.com

Vidó Gábor Bence
Department of Computer Science
Eötvös Loránd University
Pázmány Péter Sétány 1/C Budapest, Hungary
Budapest
Hungary
vidobence@gmail.com

Preszter Balázs
Department of Computer Science
Eötvös Loránd University
Pázmány Péter Sétány 1/C Budapest, Hungary
Budapest
Hungary
email

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